Start coding or generate with AI.

step 1: Install and import Libraries

```
# install NLTK (run once in colab)
!pip install NLTK
# import required modules
import nltk
from nltk.tokenize import TweetTokenizer
from nltk.corpus import twitter_samples
```

```
Requirement already satisfied: NLTK in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from NLTK) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from NLTK) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from NLTK) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from NLTK) (4.67.1)
```

step 2:Download Required NLTK Resources

```
# Download datasets and models
nltk.download ('twitter_samples')
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package twitter_samples to /root/nltk_data...
[nltk_data]   Package twitter_samples is already up-to-date!
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
True
```

step 3:Load tweets Dataset

```
# Load sample tweets
tweets=twitter_samples.strings('positive_tweets.json')
# Display sample tweets
for i in range(3):
  print("Tweet",i+1)
  print(tweets[1])
  print()
```

```
Tweet 1
@Lamb2ja Hey James! How odd :/ Please call our Contact Centre on 02392441234 and we will be able to assist you :) Many thanks!

Tweet 2
@Lamb2ja Hey James! How odd :/ Please call our Contact Centre on 02392441234 and we will be able to assist you :) Many thanks!

Tweet 3
@Lamb2ja Hey James! How odd :/ Please call our Contact Centre on 02392441234 and we will be able to assist you :) Many thanks!
```

step 4:Tokenization Using TweetTokenizer

```
# Tweettokinzer handles emojis,hashtags,abbrevations
tokinzer = TweetTokenizer(
    preserve_case=False,
    strip_handles=True,
    reduce_len=True
)
# tokenize first 5 tweets
tokinzed_tweets = [tokinzer.tokenize(tweet) for tweet in tweets[:5]]
# Display tokens
for i, tokens in enumerate(tokinzed_tweets):
  print("Tweet", i+1, "Tokens:")
```

```
    print(tokens)
    print()
```

```
Tweet 1 Tokens:
['#followfriday', 'for', 'being', 'top', 'engaged', 'members', 'in', 'my', 'community', 'this', 'week', ':)']

Tweet 2 Tokens:
['hey', 'james', '!', 'how', 'odd', ':/', 'please', 'call', 'our', 'contact', 'centre', 'on', '02392441234', 'and', 'we', 'will'

Tweet 3 Tokens:
['we', 'had', 'a', 'listen', 'last', 'night', ':)', 'as', 'you', 'bleed', 'is', 'an', 'amazing', 'track', '.', 'when', 'are', 'y

Tweet 4 Tokens:
['congrats', ':)']

Tweet 5 Tokens:
['yeaaah', 'yipppy', '!', '!', '!', 'my', 'accnt', 'verified', 'rqst', 'has', 'succeed', 'got', 'a', 'blue', 'tick', 'mark', 'on
```

step 5:

step 5: POS tagging using NLTK

```
# apply POS tagging
pos_tagged_tweets = [nltk.pos_tag(tokens) for tokens in tokenized_tweets]
#Display POS tags
for idx, tagged_tweet in enumerate(pos_tagged_tweets):
  print("Tweet", idx+1)
  print(tagged_tweet)
  print()
```

```
Tweet 1
[('#followfriday', 'NN'), ('for', 'IN'), ('being', 'VBG'), ('top', 'JJ'), ('engaged', 'VBN'), ('members', 'NNS'), ('in', 'IN'),

Tweet 2
[('hey', 'NN'), ('james', 'NNS'), ('!', '.'), ('how', 'WRB'), ('odd', 'JJ'), (':/', 'JJ'), ('please', 'NN'), ('call', 'VB'), ('o

Tweet 3
[('we', 'PRP'), ('had', 'VBD'), ('a', 'DT'), ('listen', 'VBN'), ('last', 'JJ'), ('night', 'NN'), (':)', 'NN'), ('as', 'IN'), ('y

Tweet 4
[('congrats', 'NNS'), (':)', 'VBP')]

Tweet 5
[('yeaaah', 'NN'), ('yipppy', 'JJ'), ('!', '.'), ('!', '.'), ('!', '.'), ('my', 'PRP$'), ('accnt', 'JJ'), ('verified', 'VBN'), (
```

step 6: POS Tagging on Custom Noisy Text

```
# Example informal text
text = "OMG I luv this phone 😍"#awesome #AI"
# Initialize TweetTokenizer
tokenizer = TweetTokenizer()
# tokenize
token = tokenizer.tokenize(text)
# pos tag
tags = nltk.pos_tag(token)
# print
print(token)
print(tags)
```

```
['OMG', 'I', 'luv', 'this', 'phone', '😍']
[('OMG', 'NNP'), ('I', 'PRP'), ('luv', 'VBP'), ('this', 'DT'), ('phone', 'NN'), ('😍', 'NN')]
```

step 7:Extract Nouns and Verbs

```
# Extract nouns and verbs
nouns=[]
verbs=[]
for word, tag in tags:
  if tag.startswith('NN'):
    nouns.append(word)
  elif tag.startswith('VB'):
    verbs.append(word)
```

```
print ("Nouns: ", nouns)
print ("Verbs: ", verbs)
```

```
Nouns:  ['OMG', 'phone', '😍']
Verbs:  ['luv']
```

Double-click (or enter) to edit

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```
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```

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